

Rouge Construction Network Topology

College of Science, Engineering, and Technology, Grand Canyon University

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Rouge Construction is a medium-sized business specializing in providing clients with construction support. As the company makes efforts to expand to the entire state of Kentucky it is best to develop a network infrastructure that allows for easy expansion as the network will need to grow to meet the demands of more client, financial, service, and administrative data. Some of the company functions that need to be considered when developing the network include, scheduling, logging, equipment/machinery services, financial data, supply chain management, warehouse management, HR, secure authentication, customer data management, and administrative services, and are broad enough to suit over 400 employees.

Considering the tasks that need to be performed, supported wireless mesh networks seem suitable for Rouge Construction. This topology allows for collaborative, ensured data security and backup technology and is known for its reliable, dynamic speed and performance for employees and clients within the network. With goals of expansion, this topology uses less power, is scalable, is great for temporary locations such as construction sites, is excellent for security/monitoring purposes for warehouses or machinery, and allows for public access to incoming client or vendor meetings. The initial installation is cost-intensive but is perfect for an expanding business because it will eventually save money when adding/removing devices. Within the mesh network, there should be a level 4 DMZ design that allows for separate firewall stacking within business services and user services. In total there should be 3 firewalls installed. The first will coincide with the modem so that packets that derive from the Internet are filtered prior to entering the private network. The second will team up with the router responsible for customer services/data and logging for an additional layer of security to be ensured since customer care and privacy is one of the top priorities of Rouge Construction. The third firewall

will be placed along with the router responsible for Administration devices/clients since most of the data and packets that will be passing throughout that network are sensitive and need an extra layer of protection.

